



Fakultät Informatik

From CAD Software to Slicer

Workflow Optimization using Print Hints in 3MF Files for 3D Printing

Bachelor Thesis Exposé

im Studiengang Medieninformatik

vorgelegt von

Luise Hartdegen

Matrikelnummer 364 2168

Erstgutachter: Prof. Dr. Ulrich von Zadow

Abgabetermin: angestrebt: Mitte August

© 2025

Dieses Werk einschließlich seiner Teile ist **urheberrechtlich geschützt**. Jede Verwertung außerhalb der engen Grenzen des Urheberrechtsgesetzes ist ohne Zustimmung des Autors unzulässig und strafbar. Das gilt insbesondere für Vervielfältigungen, Übersetzungen, Mikroverfilmungen sowie die Einspeicherung und Verarbeitung in elektronischen Systemen.

Contents

1	Introduction	1
2	Description	2
2.1	Definition of Problem	2
2.2	Research topic	2
2.3	Solution approach	2
2.4	Research methods	3
2.5	Current state of research	3
3	Planned timeline	4

1 Introduction

Despite automation and optimization having been a prominent topic in the software developer community for years, there still are a few workflows in 3D printing that have the potential to become even more efficient and user friendly. One of these is the process of importing 3MF files containing objects with multiple materials or modifiers to a slicer to create the print file. Oftentimes designing a 3D model for printing is an iterative process. The designer tweaks the model, prints it on their 3D printer, checks for improvement and repeats the same steps again. When doing this, the slightly altered models are being imported to the slicer again and again, meaning the designer has to set materials or e. g. mark a certain object as a support blocker repeatedly following each slicer import.

The goal of this thesis is to prove the concept of so called print hints as a way to improve the workflow. The approach as currently planned contains on the one hand an improved export function within the CAD software *Autodesk Fusion* to mark certain objects as modifiers and on the other hand automatic handling of materials and print hints within the slicer software called *OrcaSlicer*.

2 Description

2.1 Definition of Problem

When designing and printing 3D models, there are a few steps being repeated over and over again. Usually the process consists of creating the 3D model in a CAD software like Autodesk Fusion and exporting it as an STL or 3MF file. Next, this file is imported to a slicer software, where the printing parameters are being set. Since these parameters are not part of the exported file, the user has to potentially set them each time following modifications of the 3D model and reimporting it to the slicer. The printing parameters not only include the filaments, but also the surface that the object is placed on and possibly whether certain objects are to be handled as modifiers to the main 3D model.

2.2 Research topic

The goal of this bachelor thesis is to prove whether it is possible to export ‘print hints’ within the 3MF file from Autodesk Fusion and have the slicer automatically parse them to make the working process described above more efficient and avoid repetitious manual steps.

2.3 Solution approach

Based on the current knowledge, the solution will consist of multiple parts.

1. It shall be possible to add print hints within Autodesk Fusion. Since this software is not open source, the only way to add this functionality would be as a script or plugin. Autodesk Fusion provides a comprehensive amount of methods within their API for developers to use.
2. The slicer shall then be able to detect the different materials from exported 3MF files and automatically map them to filament presets, as well as automatically handle print hints.

3. Furthermore it needs to be determined, whether the 3MF standard would need to be augmented or if the print hints can be written to the xml data without changing the file standard.

2.4 Research methods

Since the goal is a proof of concept, the methods used will mainly be to create software prototypes. Optionally it might be useful to conduct user surveys as to the usability of the created prototypes. Accompanying these, the written part shall illustrate the design decisions as well as any other important steps, serve as documentation and give the reader foundational knowledge regarding the design process for 3D models and the working of slicer software.

2.5 Current state of research

Insights so far:

- Fusion exports materials as hexcode colours to the 3MF file where they are being referenced in each 3D object but not imported by the OrcaSlicer yet
- 3MF saves 3D models in easily human-readable XML format which is therefore easily edited
- OrcaSlicer doesn't throw any errors when importing manually edited 3MF files
- Fusion provides a comprehensive API to use for plugins or scripts

The most important literature will certainly be Fusion's API documentation, as well as the documentation of the OrcaSlicer and possibly the predecessors Slic3r, Bambu Studio and PrusaSlicer as far as available and up to date. Further literature is yet to be researched.

3 Planned timeline

Until	Milestone
end of May	literature research regarding the working of slicers code analysis of the OrcaSlicer implementation: automatically mapping material presets to material colours in 3MF file import first draft of the introduction
end of June	plugin to add flags to the 3MF file when exporting objects from Fusion flag handling within OrcaSlicer first draft of theoretical foundation
end of July	first complete draft
end of August	revisions, finalizations and quality control